

Centralized Treasury Take-Home — Question 2

A rich/cheap deployment-timing signal, built from the Question 1 model

Albert Wong | Walk-forward signal + backtest vs. naive even deployment, 2021-04 to 2025-12

Bottom Line

Reallocating capital deployment toward days the Question 1 model flags as **rich** (and away from **cheap** days), while holding total capital deployed constant, captured an average of **+5.6bp** more spread than deploying evenly — a **77% monthly hit rate** and **+76bp cumulative advantage** over the 57-month out-of-sample backtest. The signal also passes a basic sanity check: realized spread is monotonic across regimes (cheap 34bp → neutral 43bp → rich 60bp), confirming it carries real information rather than noise.

Signal Definition

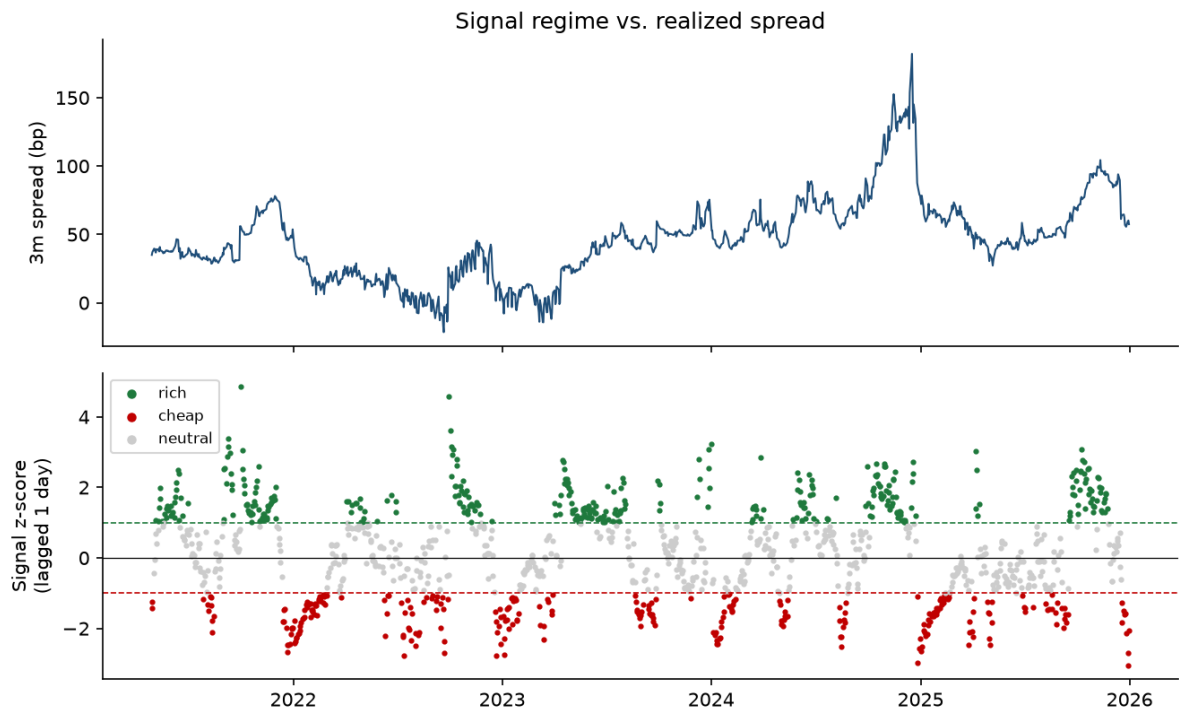
Step 1 — walk-forward fair value. The Question 1 regression is re-estimated on an expanding window, refit every 21 trading days (~monthly). Each refit uses only data strictly prior to the refit date, then is applied out-of-sample until the next refit. This matters: using the full-sample model (fit on all 5 years) to judge whether the spread was “rich” back in 2021 would be lookahead bias — those coefficients were estimated partly using 2025 data that didn't exist yet.

Step 2 — residual. $\text{residual}_t = \text{actual spread}_t - \text{walk-forward fair value}_t$.

Step 3 — z-score. The residual is standardized using a **trailing 60-day** window of past residuals only (no full-sample centering, which also evolves over time as volatility regimes change).

Step 4 — decision lag. The regime used to decide day t 's deployment is the z-score as of **day $t-1$'s close** — the decision never uses same-day information.

Thresholds: $z > +1.0 \rightarrow \text{rich}$ (deploy aggressively). $z < -1.0 \rightarrow \text{cheap}$ (delay). Otherwise neutral.



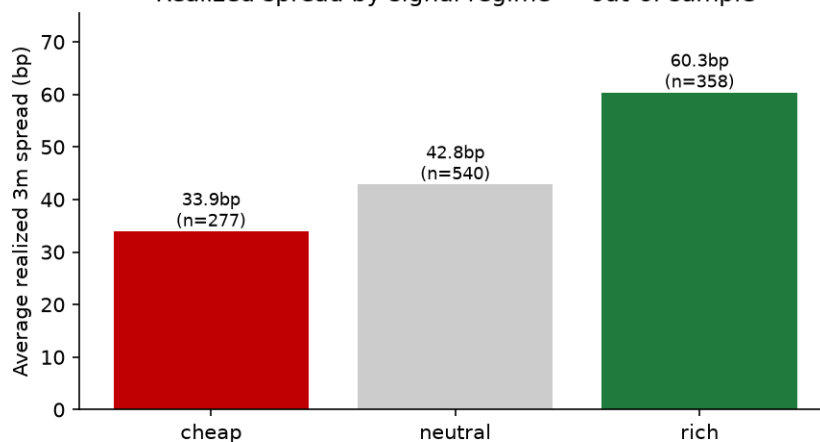
Top: realized 3m spread. Bottom: signal z-score (lagged 1 day), colored by regime. Rich/cheap clusters visibly precede the spread's own subsequent swings.

Backtest Design

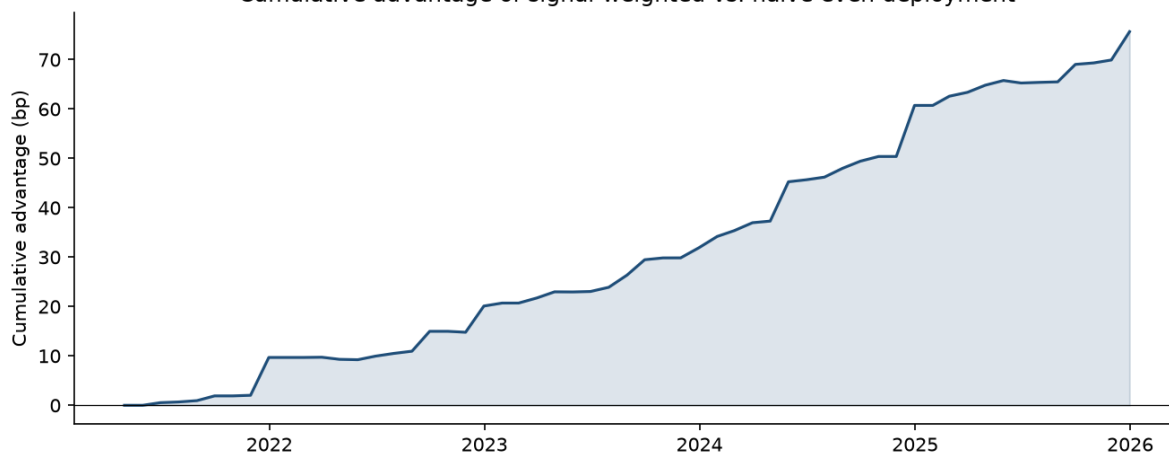
Naive baseline: deploy an equal amount each day (equivalent to capturing the simple average spread over the period). Signal strategy: reallocate the *same total capital* within each period using weights — **2.0×** on rich days, **1.0×** neutral, **0.3×** on cheap days (mostly, not fully, deferred — operationally you rarely have unlimited flexibility to defer cash indefinitely). Because both strategies deploy the same total capital, the difference in captured spread is attributable purely to *timing*, not to taking more or less risk overall.

Metric	Value
Naive average captured spread	46.05 bp
Signal-weighted average captured spread	51.63 bp
Advantage from timing	+5.58 bp
Monthly hit rate	77.2% (44 / 57 months)
Average monthly advantage	+1.33 bp
Cumulative advantage (57 months)	+75.7 bp

Realized spread by signal regime — out-of-sample



Cumulative advantage of signal-weighted vs. naive even deployment



The advantage accrues steadily across the full 4.75-year out-of-sample window rather than being concentrated in one lucky stretch — only 6 of 57 months underperformed naive, and the worst monthly shortfall was -0.51bp .

Limitations & Assumptions

Lookahead bias is addressed structurally (walk-forward refit, trailing-only z-score, 1-day decision lag) rather than just asserted — but the burn-in period (first ~14 months) had to be discarded to get a stable initial model, which is itself a modeling choice.

Weights (2.0 / 1.0 / 0.3) are a stated assumption, not fit to the data. They were chosen to be a moderate, defensible tilt rather than an aggressive one. Results are a magnitude question, not a sign question — more aggressive weights would mechanically show a larger advantage, but the monthly hit rate (which doesn't depend on the weight magnitude as strongly) is the more robust number to anchor on.

Transaction costs are not modeled. Shifting deployment day-to-day (rolling TRS/repo positions earlier or later) has real operational friction and counterparty negotiation costs that would eat into the ~5.6bp average edge.

Dealer capacity constraints work against the strategy exactly when it matters most. “Rich” days often coincide with stressed/volatile conditions (per the Q1 finding) — precisely when dealers are most likely to be balance-sheet constrained and least able to actually take the other side of the trade at the rate the signal suggests. The backtest assumes you can always get filled; live implementation cannot assume that.

Model risk carries over from Question 1: the underlying regression's relationships were shown to be partly regime-level co-movement rather than tight daily causation (see the first-difference robustness check in the Q1 report). If those macro relationships shift structurally going forward, the walk-forward refit will adapt, but with a lag — there will always be a transition period where the “fair value” is anchored to a stale regime.

Residual autocorrelation means consecutive rich/cheap days are not independent bets — the 57 monthly observations are a more honest count of independent trials than the 1,175 daily observations would suggest.